Optimal WTI—FX Hedge Ratios Under a Fixed Budget

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Abstract

A Korean crude-oil importer facing correlated WTI and USD/KRW exposure must decide, before any option is bought, what fraction of each to hedge out of a fixed spending authority, a static allocation problem this paper poses and solves as a convex program. That authority is applied to premium plus a one-point stress loss rather than to cash alone, a distinction Section 3.3 sets out and which matters for reading the result; on that ledger it binds at a vertex in the vanilla (European) regime, though only shallowly, and under a premium-only reading it binds in neither regime; under barrier (American) pricing the FX leg is cheap enough that the budget is slack and the allocation line alone pins the split, so the budget's role is regime-specific and not universal. Either way the allocation's extreme asymmetry follows in closed form from the WTI leg's dominant variance and near-zero cross-asset correlation; and cost minimization alone selects a different, worse-hedged corner than risk minimization. The paper then quantifies, and resolves, a limitation specific to knock-out pricing: its premium is cheap because the option can die, and under the stress scenario the hedge is bought against, jump-adapted exact simulation shows it is dead on roughly half the paths that reach that scenario, above the unconditional statistics a naive reading would suggest, enough to make an all-knock-out book infeasible under the very stress it targets. An instrument-mix program letting the optimizer choose between a standalone-priced knock-out leg and a vanilla leg resolves this: feasible at every mortality level, it switches instruments at a closed-form indifference point and, at the measured stress mortality, selects the all-vanilla book, budget-feasible at no cost in residual volatility against the knock-out program's own optimum. The knock-out structure's premium discount, at its measured mortality, is decisively not worth having. The reported split is conditional on two modelling choices whose consequences the paper traces rather than assumes: a calm-sample correlation estimate, and a coverage cap under which the two legs share one notional envelope.

Keywords: corporate hedging; hedge ratio; convex optimization; knock-out barrier; quanto; commodity and currency risk; budget constraint

JEL classification: G32; G13; Q40; C61

Contents

1 Introduction

Before a hedging desk ever debates *how* to run an option hedge (which pricing model, which rebalancing rule, which Greeks) the treasurer has to answer a cruder and more consequential question: out of a fixed hedging budget, *what fraction of each exposure should be covered at all?* For a Korean crude-oil importer the exposures are two: the WTI price of the physical barrels it must buy every month, and the USD/KRW exchange rate at which the resulting dollar liability is converted into won. In the configuration studied here the physical requirement is 2,000,000 barrels per month, the equivalent dollar need is \$157,880,000 per month, and the approved monthly option budget is KRW 45,000,000,000. The decision variables are two scalars: $w_1 \in [0, 1]$, the fraction of the WTI exposure covered by WTI call options, and $w_2 \in [0, 1]$, the fraction of the USD need covered by USD/KRW call options. This is a *static* allocation made once, before the hedge is put on; nothing in it depends on how the position is subsequently managed.

The economics of the trade-off are stark on both sides. Hedging is expensive: covering the full oil leg with Black-76-priced calls costs KRW 41.3bn (nearly the entire budget), and covering the full FX leg costs KRW 13.8bn with the Garman—Kohlhagen premium or KRW 4.2bn with the American structure's Shapley-attributed currency premium. Not hedging is also expensive: under the stress scenario against which the program is sized (WTI at 113 USD/bbl, USD/KRW at 1550), the uncovered oil exposure alone loses KRW 105.6bn if nothing is hedged. The allocation problem is the collision of these two ledgers under one spending cap.

Three questions organize the paper, and their answers are its results. First, **is the budget constraint actually binding**: is the optimum a boundary point held back by money, or an interior risk optimum that merely happens to sit near the cap? Second, **what mechanism produces the extreme w_1/w_2 asymmetry** the solutions exhibit: is it an artifact of the optimizer, or a structural consequence of the volatility and correlation inputs? Third, **would minimizing cost directly give the same allocation** as minimizing risk subject to the budget, and if not, which criterion should govern? Two further analyses address a property specific to the American regime: a knock-out option's premium is low precisely because the option can die, and the scenario in which protection is most needed is also the scenario in which death is most likely. The paper re-simulates the calibrated jump-diffusion from the stress state itself to estimate that survival risk directly, and not quoting unconditional knock-out statistics, and then prices it into the allocation through a mixed vanilla/knock-out program in which the survival-adjusted ledger is the budget constraint and the optimizer chooses the instrument as well as the coverage.

Both allocation programs are convex, so their global solutions can be characterized in closed form and verified numerically. The answers, previewed: (a) whether the combined premium-plus-stress ledger binds is regime-specific: in the European regime it binds at a vertex (the ledger cap and the allocation envelope hold with equality simultaneously) but shallowly, in the sense that unlimited budget buys almost no further variance reduction, while in the American regime the cap is slack and the allocation envelope binds alone; (b) the asymmetry is structural and can be written in one closed-form line, with the FX leg left 96.6% open by the variance arithmetic itself; and (c) cost minimization is a different program with a different (corner) answer, and the two criteria coincide only when the cost program is re-anchored by the risk cap it discards.

2 Literature Review

The objective function is the classical minimum-variance portfolio criterion of Markowitz (1952), specialized to two residual exposures: the quantity minimized is the standard deviation of a two-asset portfolio whose weights are the *unhedged* fractions $(1-w_1)$ and $(1-w_2)$. The closed-form line-restricted solution used in Section 6.2 to explain the allocation asymmetry is the textbook two-asset global-minimum-variance weight.

The premium inputs on the European side come from two closed-form option models. Black (1976) prices the WTI leg as a call on the commodity forward; Garman and Kohlhagen (1983) price the FX leg as a currency call with separate domestic and foreign interest rates. On the American side, premiums come from a Longstaff—Schwartz (2001) least-squares Monte Carlo valuation of a knock-out structure under jump-diffusion dynamics in the spirit of Merton (1976); the diffusive volatility 0.3242 that valuation consumes is what an expectation—maximization decomposition leaves after stripping the jump variance out of the raw historical volatility 0.3946, and it is an input to the pricing engine and not to the risk objective, which takes the raw figure because the uncovered barrels bear the jumps as well as the diffusion. The joint quanto premium is never assembled from a closed-form correlation-drift adjustment of the kind derived by Reiner (1992); it is instead recovered directly from the correlated joint simulation of Section 4, which reproduces the same coupling automatically once S_1 and S_2 are drawn from correlated shocks. Because the American structure prices both legs jointly, the joint premium is attributed to the individual legs by Shapley (1953) value, producing per-leg premium constants that enter the cost function linearly. The premiums are priced at inception, at today's spot, deliberately so: the budget constraint governs cash paid *today*, and re-pricing the premium at the stress spot would charge the stress scenario twice, once in the premium and once in the unhedged-loss terms that already price it.

The economic question (how much of each of two correlated exposures to cover when hedging is costly) descends from the cross-hedging literature of Anderson and Danthine (1981) and, for the currency dimension of internationally sourced commitments, Adler and Dumas (1984). The present setting differs from the classical futures cross-hedge in one decisive respect: hedging here is done with options bought under a hard spending authority, so the operative trade-off runs between variance and an explicit cap on a combined premium-plus-stress ledger, in place of the usual variance/expected-return frontier, which is precisely what turns a one-line minimum-variance formula into a constrained program worth studying. The unconstrained special case of that formula is nonetheless the classical minimum-variance hedge ratio of Ederington (1979); the budget cap of Section 3.3 is what separates the present, corner- and vertex-seeking program from a direct application of that one-line result.

The premium budget is taken here as exogenous, following the corporate hedging literature that supplies its rationale. Froot, Scharfstein, and Stein (1993) model this kind of institutional friction: a firm hedges to protect its internal financing capacity for investment and not to maximize variance reduction per se, which is consistent with hedging being bounded by an approved cash outlay and not pursued to a first-order risk optimum. Stulz (1996) develops the practitioner-facing corollary (selective, partial hedging can be optimal even when full coverage is affordable), which is the same qualitative conclusion Section 6.2's closed-form asymmetry reaches by a different, purely variance-arithmetic route. Jin and Jorion (2006) supply the closest empirical anchor to the present institutional setting, documenting derivative-based hedging behavior among U.S. oil and gas producers; the importer studied here sits on the opposite side of the same commodity risk, but the same budget-bounded, partial-coverage pattern they document empirically is what Sections 6—8 derive from first principles for this specific instrument set.

3 The Static Allocation Problem

3.1 Decision Variables and Risk Objective

Let σ_1 denote the annualized volatility of WTI log returns, σ_2 that of USD/KRW log returns, and ρ their correlation. If fractions w_1 and w_2 of the two exposures are hedged, the unhedged remainders $(1-w_1)$ and $(1-w_2)$ carry the residual risk

$$\sigma_{res}(w_1, w_2) \;=\; \sqrt{(1-w_1)^2\sigma_1^2 \;+\; (1-w_2)^2\sigma_2^2 \;+\; 2(1-w_1)(1-w_2)\,\sigma_1\sigma_2\rho}\,. \tag{1}$$

Both regimes share every risk input. σ_{res} in (1) measures the volatility of the exposure left uncovered, and that exposure does not know which model priced its hedge, so $\sigma_1 = 0.3946$, $\sigma_2 = 0.0926$ and $\rho = 0.0876$ enter it whichever instrument is bought. The jump-stripped diffusive figure $\sigma_1^{diff} = 0.3242$ appears only inside the American Monte Carlo engine, which restores the jump component separately; taking it into the risk objective would credit the uncovered barrels with a jump immunity they do not have.

3.2 Cost Functions

Total cost is the sum of two ledgers: the premium paid today for the covered fraction, and the stress-scenario loss suffered by the uncovered fraction. Both regimes share the stress ledger and differ only in the premium ledger. With $Q_{oil} = 2{,}000{,}000$ bbl, $Q_{USD} = \$157{,}880{,}000$, spot prices $S_{WTI} = 78.94$ and $S_{KRW} = 1540.64$, stress levels $S_{WTI}^{st} = 113$ and $S_{KRW}^{st} = 1550$, funding rate $r_w = 0.07$ (WACC), and maturities $T_1 = 0.833$, $T_2 = 0.5$ years:

$$\begin{aligned} C^{EU}(w_1, w_2) \;=\;& w_1\, Q_{oil}\, P_{B76}\, S_{KRW}\,(1 + r_w T_1) \;+\; w_2\, Q_{USD}\, P_{GK}\,(1 + r_w T_2) \\ &+\; (1-w_1)\, Q_{oil} \max(0,\, S_{WTI}^{st} - S_{WTI})\, S_{KRW}^{st} \\ &+\; (1-w_2)\, Q_{USD} \max(0,\, S_{KRW}^{st} - S_{KRW}), \end{aligned} \tag{2}$$

$$C^{AM}(w_1,w_2) \;=\; w_1\,Q_{oil}\,P^{Sh}_{WTI}\,(1+r_wT_1) \;+\; w_2\,Q_{oil}\,P^{Sh}_{FX}\,(1+r_wT_2) \\ +\;(1-w_1)\,Q_{oil}\max(0,\,S^{st}_{WTI}-S_{WTI})\,S^{st}_{KRW} \\ +\;(1-w_2)\,Q_{USD}\max(0,\,S^{st}_{KRW}-S_{KRW}), \tag{3}$$

with $P_{B76} = 12.6524$ USD/bbl (Black-76 WTI call), $P_{GK} = 84.667$ KRW per USD (Garman—Kohlhagen FX call), and $P^{Sh}_{WTI} = 15{,}093.75$, $P^{Sh}_{FX} = 2{,}038.72$ KRW/bbl (LSMC Shapley attribution of the joint per-barrel premium 17,132.47 KRW/bbl, so both shares scale on the barrel position). Both functions are affine, and collecting terms gives the marginal cost of each ratio:

$$\frac{\partial C^{EU}}{\partial(w_1,w_2)} = (-64.327,\ +12.357)\text{ bn KRW}, \qquad \frac{\partial C^{AM}}{\partial(w_1,w_2)} = (-73.586,\ +2.745)\text{ bn KRW}. \tag{4}$$

The signs carry the economics. Raising the WTI ratio *reduces* total cost in both regimes: the avoided stress loss (KRW 105.59bn at full exposure) dwarfs the full-cover premium (KRW 41.26bn European, KRW 32.00bn American). Raising the FX ratio *increases* it: in both regimes the FX premium exceeds the small FX stress gap (KRW 1.48bn at full exposure), by KRW 12.36bn per unit of w_2 in the European regime and KRW 2.75bn in the American, where the barrier structure prices the currency leg far more cheaply. Equation (4) drives everything in Section 6.3.

3.3 Constraint Set

The allocation program imposes exactly four constraints:

$$0 \le w_1 \le 1, \qquad 0 \le w_2 \le 1, \qquad w_1 + w_2 \le 1, \qquad C(w_1,w_2) \;\le\; B = 45{,}000{,}000{,}000\text{ KRW}, \tag{5}$$

with C equal to (2) or (3).

What B actually caps. One point must be made before any result is read, because the paper's earlier drafts described the constraint incorrectly. $C(w_1, w_2)$ in (5) is not cash paid at inception. It is the sum of the option premium, which is cash, and the stress-scenario loss on the unhedged fraction, which is a contingent outcome under one scenario. At the European risk-minimising vertex the two components are KRW 40.45bn of premium and KRW 4.55bn of stress loss, summing to the KRW 45bn cap. Under a cap on *premium alone* the same allocation spends 40.45bn, the unconstrained line optimum (0.9660, 0.0340) spends 40.33bn, and the cap binds in neither regime.

The vertex result of Section 6.1 is therefore a statement about a combined spend-plus-stress ledger and not about a cash authority, and the reader should hold that distinction throughout. A treasury whose 45bn authority governs cash outlay only would find both regimes interior and the allocation pinned by (7) alone, which is the American answer (0.9660, 0.0340) in both cases. The paper reports the combined-ledger program because that is what makes the two regimes' stress exposure comparable on one axis, but the "budget binds" language of question (a) applies to that ledger and not to a premium budget.

The allocation constraint $w_1+w_2 \le 1$ deserves more than a passing note, because the paper's central asymmetry result is derived on the line it defines. Its economic content is that the two legs are denominated against *the same* exposure. At the spot rate of Table 1 the monthly barrel requirement and the monthly dollar requirement are the same quantity of money,

$$Q_{oil} \cdot S_{WTI} \;=\; 2{,}000{,}000 \times 78.94 \;=\; \$157{,}880{,}000 \;=\; Q_{USD}, \tag{6}$$

so a barrel hedged and a dollar hedged cover the identical underlying cash outflow, once through its price and once through its currency of settlement. A treasury that authorizes hedging of "the monthly crude bill" and treats w_1 and w_2 as two ways of covering that one bill is imposing precisely $w_1 + w_2 \leq 1$. Read this way the constraint is a statement about what is being hedged rather than an arbitrary cap.

The reading is not forced, however, and the results are sensitive to it. A firm that regarded the price exposure and the currency exposure as two separate books, each hedgeable up to its own notional, would impose $w_1 + w_2 \leq \kappa$ with $\kappa = 2$, and Section 6.6 shows what that does to the answer. The constraint is therefore reported here as a modelling choice with an economic rationale, and its consequences are traced rather than assumed away.

Together with the budget cap these constraints carve the feasible polygon whose geometry Section 6 maps completely: a sliver satisfying $w_1 \geq 0.9648 + 0.1921\, w_2$ in the European regime, and a far roomier, nearly horizontal band satisfying $w_1 \geq 0.8434 + 0.0373\, w_2$ in the American regime.

4 Data and Calibration

Table 1 lists every input used in the paper. The underlying series are daily WTI spot (USD/bbl) and USD/KRW, covering 28 June 2021 to 22 June 2026, 1,301 paired closing observations retained after aligning the two calendars and dropping non-common trading days. Differencing gives 1,300 paired daily log returns, of which 1,299 survive the jump-classification screen of Section 7 and constitute the estimation sample used throughout. The risk inputs are computed from those 1,299 paired daily log returns: $\sigma_1 = 0.3946$ (WTI, annualized) and $\sigma_2 = 0.0926$ (USD/KRW), with correlation $\rho = 0.0876$. The American regime's $\sigma_1 = 0.3242$ is the diffusive volatility from the expectation—maximization jump/diffusion decomposition of the same series. The two closed-form legs are struck 5% in the money, at $0.95\, S_{WTI} = 74.99$ USD/bbl and $0.95\, S_{KRW} = 1{,}463.61$ KRW/USD; the American structure is struck at the money, $K = S_{WTI} = 78.94$, with knock-out barriers at 120 and 50 USD/bbl. The two regimes therefore price different strikes, which is worth holding in view wherever their cash ledgers are compared side by side. The stress scenario (WTI at 113 USD/bbl and USD/KRW at 1550) prices the unhedged fraction of each leg in (2)—(3).

5 Numerical Method and Verification

Both allocation programs are convex: σ_{res} in (1) is the Euclidean norm of an affine map of (w_1, w_2), hence convex; the cost functions (2)—(3) are affine; and every constraint in (5) is affine. Any local minimum is therefore the global minimum, and the solution can be established

Quantity	Symbol	Value
WTI spot (USD/bbl)	S_{WTI}	78.94
USD/KRW spot	S_{KRW}	1,540.64
Monthly oil need (bbl)	Q_{oil}	2,000,000
Monthly USD need	Q_{USD}	157,880,000
WACC (funding rate)	r_w	0.07
Budget (KRW)	B	45,000,000,000
European WTI call strike (USD/bbl)	$K_1 = 0.95\, S_{WTI}$	74.99
European FX call strike (KRW/USD)	$K_2 = 0.95\, S_{KRW}$	1,463.61
American call strike (USD/bbl)	$K = S_{WTI}$	78.94
Knock-out barriers (USD/bbl)	U, L	120, 50
WTI option maturity (yr)	T_1	0.833
FX option maturity (yr)	T_2	0.5
Stress WTI (USD/bbl)	S^{st}_{WTI}	113
Stress USD/KRW	S^{st}_{KRW}	1,550
WTI volatility, risk objective	σ_1	0.3946
WTI diffusive vol, pricing engine	σ_1^{diff}	0.3242
USD/KRW vol	σ_2	0.0926
WTI—FX correlation	ρ	0.0876
USD risk-free rate (pricing)	r_{US}	0.040
KRW risk-free rate (pricing)	r_{KRW}	0.035
Jump intensity (symmetrized)	λ	6.7846 /yr
Mean jump size (log)	θ_J	-0.0299
Jump volatility (log)	δ_J	0.08443
Black-76 WTI call (USD/bbl)	P_{B76}	12.6524
Garman—Kohlhagen FX call (KRW/USD)	P_{GK}	84.667
LSMC Shapley WTI premium (KRW/bbl)	P^{Sh}_{WTI}	15,093.75
LSMC Shapley FX premium (KRW/bbl)	P^{Sh}_{FX}	2,038.72

Table 1: Model inputs. The risk and cost blocks are estimated from the price history described in Section 4; the jump block and the two pricing rates enter only the American premium and the knock-out simulation of Sections 7–8. Every downstream number in this paper is computed from these values.

analytically and not only searched for.

Two complementary numerical approaches confirm the global optima. A deterministic grid search (801 points per axis, twice refined) covers the feasible region exhaustively. In parallel, a sequential least-squares quadratic programming routine (SLSQP, 73 multi-starts) recovers the boundary solutions alongside closed-form analytic derivations from the affine constraint algebra. Across all evaluated scenarios the two solvers and the analytic bounds locate the same vertices.

The convex nature of the programs also permits a duality check, exploited throughout Section 6: the cost minimum taken subject to a risk cap equal to the achieved minimum risk returns the risk-minimizing point itself. Both regimes satisfy this duality constraint, consistent with the derived allocations being the constrained optima.

Two further computations feed Sections 7 and 8. The scenario-conditional knock-out probability of Section 7 is measured by direct simulation of the calibrated WTI jump-diffusion (diffusive volatility 0.3242, jump intensity $\lambda = 6.7846$ per year, jump mean -0.0299, jump volatility 0.08443, physical drift 0.139, Merton compensator in the drift) started *at the stress spot* and monitored against the 120/50 barriers over the option's 0.833-year life with jump-adapted exact first-passage sampling: Poisson jump times are drawn exactly within each daily step, the Brownian-bridge crossing probability is applied on each jump-free sub-interval, and the barrier is checked directly at each jump instant, so the reported touch probability carries no time-discretization bias (200,000 paths; coarser monitoring understates this probability materially, since a barrier that is touched and re-crossed inside a recording gap leaves no trace in the recorded values). The mixed-instrument program of Section 8 exploits an exact reduction: its survival-adjusted ledger is linear in the vanilla/KO split at any fixed total coverage, so the optimal split is always at an endpoint (all-vanilla or all-KO), and the two-instrument program collapses to the lower envelope of two two-variable convex programs solved with the same machinery.

6 Empirical Results

Table 2 collects the six solved programs together with the two budget-relaxed benchmarks; the subsections interpret them.

Regime	Program	w_1	w_2	Total cost (KRW bn)	σ_{res}
European	risk min. (budgeted)	0.9705	0.0295	45.000	0.0916
European	risk min. (no budget)	0.9660	0.0340	45.346	0.0916
European	cost min.	1.0000	0.0000	42.737	0.0926
European	cost min. $\leq$ risk cap	0.9705	0.0295	45.000	0.0916
American	risk min. (budgeted)	0.9660	0.0340	36.075	0.0916
American	risk min. (no budget)	0.9660	0.0340	36.075	0.0916
American	cost min.	1.0000	0.0000	33.478	0.0926
American	cost min. $\leq$ risk cap	0.9660	0.0340	36.075	0.0916

Table 2: Solved allocation programs. In the European regime the budgeted risk minimum is a vertex solution: the allocation constraint $w_1 + w_2 \leq 1$ and the budget constraint hold with equality simultaneously. In the American regime the budget is slack at the optimum, so the allocation constraint binds alone and the budgeted and unbudgeted minima coincide. The risk-capped cost minima reproduce the risk minima (the duality audit of Section 5).

6.1 Question (a): Where the Budget Binds, a Vertex in One Regime and Slack in the Other

Three computations characterize the budget's role.

First, the active set. At the European risk minimum the allocation constraint and the budget constraint hold with equality *simultaneously*: $w_1^* + w_2^* = 1$ and $C(w_1^*, w_2^*) = B$. The optimum is the vertex where the budget boundary crosses the allocation line, visible in the zoomed panels of Figure 3, where the iso-risk contours close in on a point pinched between the solid budget boundary and the dashed allocation line. The American regime has a different active set: only $w_1^* + w_2^* = 1$ holds, at a total cost of KRW 36.07bn, KRW 8.93bn inside the cap, so the budget boundary runs well below the solution and never touches it.

Second, the relaxation test. Deleting the budget constraint moves the European optimum from $(0.9705, 0.0295)$ to $(0.9660, 0.0340)$, which lowers σ_{res} only slightly; funding that relaxed optimum would cost KRW 45.35bn, 0.77% over budget and so unaffordable. The cap therefore genuinely excludes the unconstrained European optimum, and the ledger constraint is binding there in the exact Karush—Kuhn—Tucker sense. The American optimum does not move at all: its unconstrained minimum $(0.9660, 0.0340)$ already costs only KRW 36.07bn, some 20% inside the cap, so the constraint is inactive and its multiplier is zero. The regimes differ because the barrier structure prices the currency leg at KRW 2.75bn per unit of w_2 against the vanilla leg's KRW 12.36bn: the FX coverage the risk objective wants is affordable in one regime and not in the other.

Third, the price of the constraint. Figure 1 traces the optimal σ_{res} as the budget sweeps from KRW 33bn to 58bn. Each curve falls, then flattens at the relaxed-optimum budget just computed (KRW 45.35bn European, KRW 36.07bn American), beyond which money buys nothing. The KRW 45bn cap sits on the falling branch of the European curve and on the flat branch of the American one. Finite-difference shadow prices at $B = 45$bn are accordingly about one basis point of σ_{res} per additional billion KRW in the European regime and zero in the American. Where it binds the constraint is moreover *shallow*: even unlimited budget improves European residual volatility by 0.02%. The treasurer's cap costs the firm almost no variance; conversely, requests for more hedging budget cannot be justified on variance grounds in this configuration.

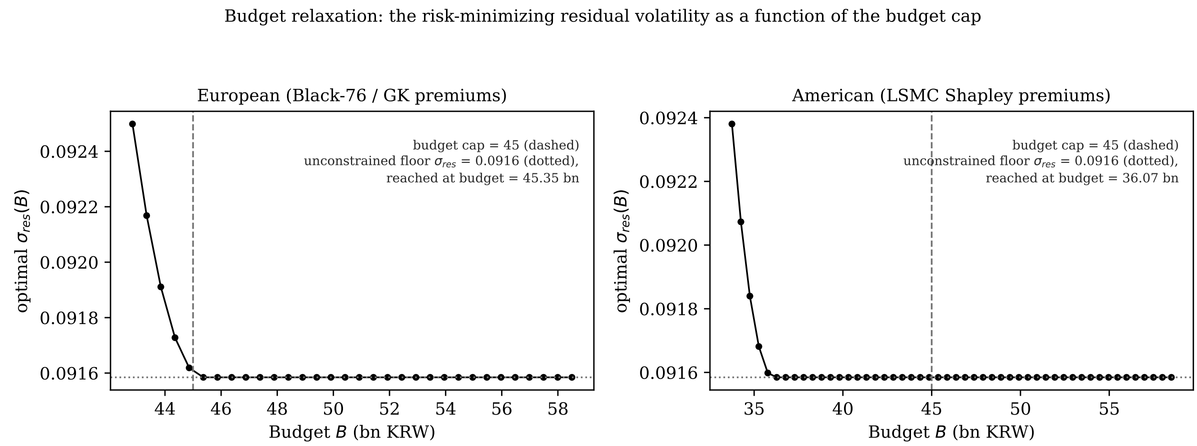

Figure 1: Budget relaxation sweep. The optimal residual volatility falls as the cap is raised and flattens at the unconstrained floor, reached at KRW 45.35bn (European) and KRW 36.07bn (American). The KRW 45bn cap (dashed) sits on the falling branch of the European curve, binding but close to the floor, and on the flat branch of the American curve, where it is slack.

6.2 Question (b): The Mechanism Behind the Extreme w_1/w_2 Asymmetry

The extreme tilt toward the WTI leg follows from three structural facts and not from any preference of the optimizer, each visible in closed form.

Fact 1: the WTI leg carries almost all the risk. $\sigma_1^2/\sigma_2^2 = 18.2$. At the no-hedge point $(0, 0)$, where $\sigma_{res} = 0.4131$, the gradient of (1) is

$$\nabla\sigma_{res}(0, 0) = (-0.3846,\ -0.0285),$$

i.e. the first won of hedging is 13.5× more productive on the WTI leg. A budget spent greedily on marginal risk reduction floods the WTI leg first.

Fact 2: the correlation is too small to couple the legs. The interaction term in (1) carries the factor $\sigma_1\sigma_2\rho = 0.0032$ (European), two orders of magnitude below σ_1^2. With $\rho = 0.0876$, hedging WTI does almost nothing to the FX leg's marginal value and vice versa: the problem nearly decouples into two independent legs competing for one budget, and the leg with 18× the variance wins almost all of it.

Fact 3: the allocation line pins the split in closed form. Because σ_{res} is strictly decreasing in both w_1 and w_2 (its gradient is negative throughout the unit square), any optimum must exhaust the binding envelope: the line $w_1 + w_2 = 1$, intersected with the budget. On that line, substituting $w_2 = 1 - w_1$ into (1) gives the classical two-asset minimum-variance problem in the unhedged weights $h_1 = 1 - w_1$, $h_2 = w_1$, whose solution is

$$w_1^{line} \;=\; \frac{\sigma_1^2 - \rho\,\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\,\sigma_1\sigma_2} \;=\; 0.9660 \qquad \text{(both regimes).} \tag{7}$$

Equation (7) *is* the asymmetry: with these inputs no optimizer on the allocation line can choose anything but $w_2 = 1 - w_1^{line} = 0.0340$. The residual FX openness is not neglect; it is the minimum-variance split of one notional envelope across an 18-to-1 variance mismatch with near-zero correlation. The budget then slides the European solution along the line to its vertex (upward in w_1, because (4) makes w_1 the cheap direction and w_2 the expensive one): from 0.9660 to 0.9705. In the American regime there is no such slide. The cap is slack by KRW 8.93bn, so the optimum simply rests at the line minimum-variance point 0.9660 that (7) delivers, and the FX leg keeps the full 0.0340 of cover the variance arithmetic asks for. The two regimes therefore share one unconstrained optimum and separate only through the budget.

6.3 Question (c): Cost Minimization Is a Different Program with a Different Answer

Because both cost functions are affine (4), pure cost minimization over the feasible polygon is a linear program whose optimum is a corner, identifiable by inspection: cost falls in w_1 and rises in w_2, so the minimum is $(w_1, w_2) = (1, 0)$ (cover the oil leg completely, leave the currency leg completely open) in *both* regimes, confirmed by the solvers and by corner enumeration. The European cost minimum spends KRW 42.74bn (5.0% under budget) and accepts $\sigma_{res} = 0.0926$, which is +1.07% relative to the risk minimum; the American cost minimum spends KRW 33.48bn (25.6% under budget) at the same $\sigma_{res} = 0.0926$ (+1.08% relative), the two regimes coinciding in risk because at $(1, 0)$ the only surviving exposure is the FX leg, whose volatility is common to both.

The answer to question (c) is therefore an unambiguous *no*: "risk minimization subject to a budget" and "cost minimization" select different points in both regimes, a solution on the allocation line versus a polygon corner. The gap between them prices the last sliver of FX coverage: moving from the cost corner $(1, 0)$ to the risk solution trades KRW 2.26bn (European) or KRW 2.60bn (American) of additional spending for a 1.1% relative reduction in residual volatility: the entire remaining decision, compressed into one trade.

Re-anchoring the cost program with an explicit risk cap closes the gap: minimizing cost subject to $\sigma_{res} \le \sigma_{res}^*$ (the achieved risk minimum) returns the risk-minimizing vertex itself in both regimes (Table 2). Economically this says the risk minima are *cost-efficient*: no cheaper allocation attains their risk level, so the budgeted risk program is already sitting on the cost—risk frontier. Methodologically it is the duality audit of Section 5 passing on live data.

6.4 The Global Geometry

Figure 2 assembles the whole problem on the full unit square: the residual-volatility surface (1) computed on a 201×201 grid, with all four constraints (5) evaluated at every node. Feasible faces are shaded (dark = low risk) and outlined in black; the constraint-violating remainder of the surface is drawn as a bare wireframe. Two properties are immediate. First, how little of the decision space survives the constraints: a sliver hugging $w_1 \gtrsim 0.965$ in the European regime, and a far roomier band rising from $w_1 = 0.843$ in the American regime, whose near-horizontal boundary slope of 0.0373 is the direct image of the KRW 4.22bn FX premium. Second, the risk minima (filled circles) sit where the shaded region touches its own darkest values: the minimum *within* the feasible shading, categorically distinct from the surface's global minimum at the infeasible corner $(1, 1)$, where $\sigma_{res} = 0$ but the cost ledgers read KRW 55.1bn (European) and KRW 36.2bn (American) and the allocation envelope is violated outright. The cost-minimizing corner $(1, 0)$ (open triangles) is visibly a different point in both panels, and the zoomed bottom row resolves what the full-square view compresses: the shaded surface patch, its black boundary, and the two optima pinned to it by their drop stems. Figure 3 gives the same geometry in contour view with a zoom: the budget boundary (solid), the allocation line (dashed), the shaded feasible polygon, and the three solution markers per panel, making the vertex structure of Section 6.1 and the corner-versus-vertex contrast of Section 6.3 visible at a glance.

6.5 Parameter Robustness

Consider the worst case over an uncertainty box. Since σ_{res} is monotone increasing in σ_1, σ_2 and ρ (all unhedged weights are nonnegative), the worst case is the upper corner of the box in closed form, and the min—max program replaces the objective with σ_{res} evaluated at inflated parameters. The feasible polygon is built entirely from prices, quantities and the budget, so it does not move; only the objective does. What happens next depends on how the box is drawn, and the distinction matters more than it first appears.

Inflating the volatilities alone leaves both allocations untouched. Scaling σ_1 and σ_2 by a common factor leaves the variance ratio σ_1^2/σ_2^2 unchanged, and (7) depends on the two volatilities only through that ratio and through ρ. The line solution therefore stays at $w_1 = 0.9660$, which lies inside the European budget boundary at $w_1 = 0.9705$, so the European optimum remains the vertex $(0.9705, 0.0295)$ and the American optimum remains $(0.9660, 0.0340)$. Only the achieved risk re-prices, from 0.0916 to 0.1008. This is the sense in which a constraint-pinned optimum is robust: it is robust to the parameters that do not move the line.

Shifting the correlation moves both. Correlation is not such a parameter. Taking the box to be both volatilities inflated by 10% *and* ρ shifted up by 0.05, to $\rho = 0.1376$, the line solution (7) moves to

$$w_1^{line}\big|_{\rho=0.1376} \;=\; 0.9770, \tag{8}$$

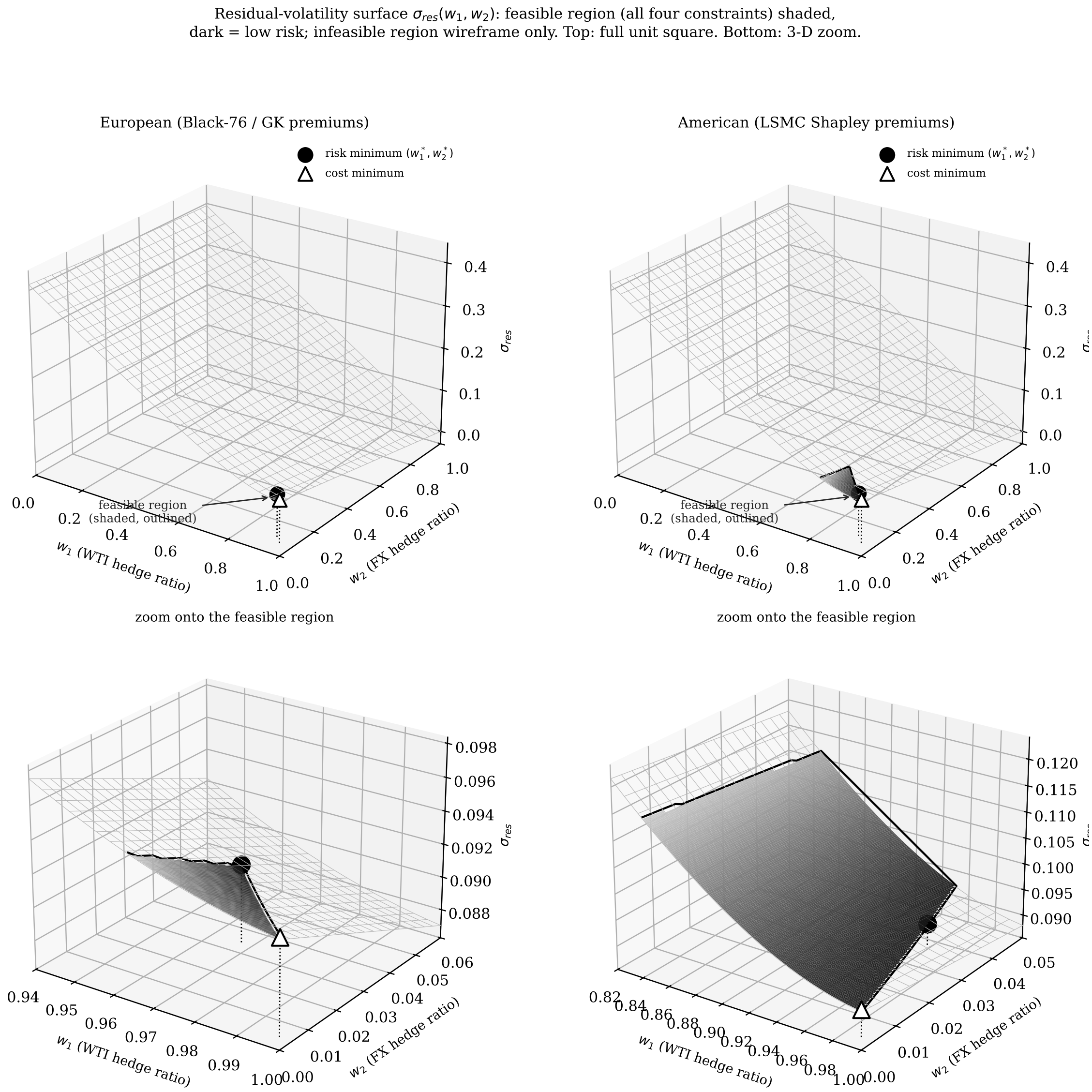

Figure 2: Residual-volatility surface over the hedge-ratio square. Shaded faces (dark = low risk, black outline) satisfy all four constraints; the wireframe region violates at least one. Top row: full unit square. Bottom row: three-dimensional zoom onto the feasible neighborhood, where the vertex structure is legible: dotted stems drop from each optimum to the panel floor. Filled circle: risk minimum. Open triangle: cost minimum $(1, 0)$.

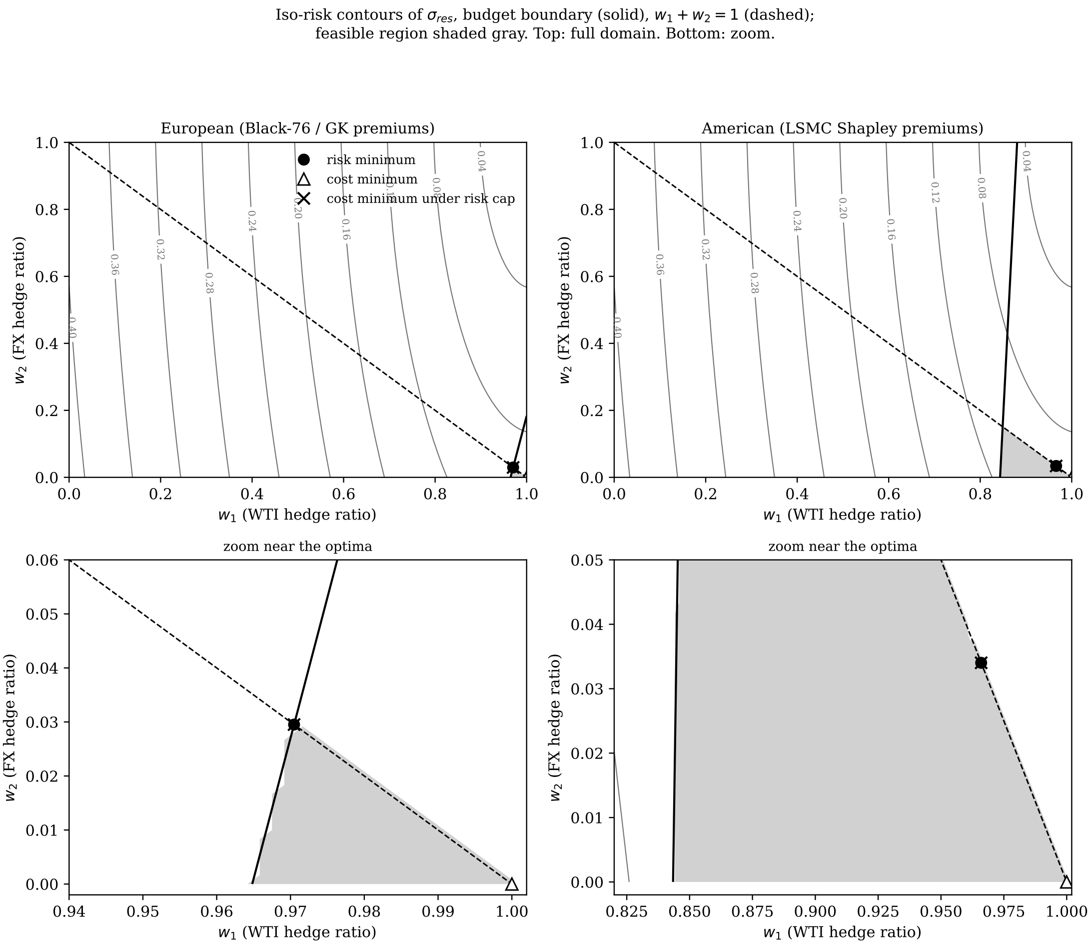

Figure 3: Contour view of the same geometry. Iso-σ_{res} contours (gray), budget boundary (solid black), allocation line $w_1 + w_2 = 1$ (dashed), feasible region (shaded). Bottom row zooms onto the optima: the risk minimum (filled circle) sits at the vertex of the feasible polygon; the cost minimum (open triangle) at the corner $(1, 0)$; the risk-capped cost minimum ($\times$) coincides with the risk minimum: the duality audit passing.

which now lies *outside* the European budget boundary $w_1 \geq 0.9705$. The point (0.9770, 0.0230) costs KRW 44.50bn, inside the KRW 45bn cap, so it is feasible and the cap goes slack. Both regimes therefore relocate to the same interior point (0.9770, 0.0230), with achieved risk 0.1013 in each. The European vertex is not preserved under this box: the budget stops binding, and the optimum is pinned by curvature in both regimes rather than by the cap in one.

The contrast is the substantive result of this section, and it sharpens the limitation stated in Section 11. Vertex-pinning delivers robustness against proportional volatility mis-estimation, which is the error mode a desk usually has in mind, because such errors preserve the variance ratio that (7) actually reads. It delivers no robustness at all against correlation error, because ρ enters (7) separately and can push the line solution across the budget boundary, at which point the constraint that was doing the pinning simply stops binding. A shift of $+0.05$ in ρ, small next to the migration documented in stressed regimes, is already enough to do this. Estimation error therefore changes what the allocation *delivers* under every perturbation considered here, and changes what the allocation *is* whenever the perturbation moves ρ.

6.6 How much of the answer is the allocation constraint? A κ-sweep

Section 6.2 attributes the extreme w_1/w_2 split to the variance arithmetic. That attribution is conditional on $w_1+w_2 \leq 1$, because (7) is derived on the line $w_1+w_2 = 1$; it describes where the minimum sits *along* that line, not whether the line is where the optimum belongs. Relaxing the cap to $w_1 + w_2 \leq \kappa$ separates the two questions. Table 3 reports the risk-minimizing program at $\kappa \in [1, 2]$, holding budget, premiums and the stress ledger fixed.

	European				American			
κ	w_1	w_2	σ_{res}	cost	w_1	w_2	σ_{res}	cost
1.00	0.9705	0.0295	0.09160	45.00	0.9660	0.0340	0.09158	36.07
1.25	1.0000	0.1831	0.07562	45.00	0.9745	0.2755	0.06869	36.11
1.50	1.0000	0.1831	0.07562	45.00	0.9830	0.5170	0.04579	36.15
1.75	1.0000	0.1831	0.07562	45.00	0.9915	0.7585	0.02290	36.19
2.00	1.0000	0.1831	0.07562	45.00	1.0000	1.0000	0.00000	36.22

Table 3: Risk-minimizing allocation as the coverage cap $w_1 + w_2 \leq \kappa$ is relaxed. Costs in KRW bn against the KRW 45bn budget. At $\kappa = 1$ the reported optima of Section 6.1 are recovered. Relaxing κ changes the answer materially, and in the American regime removes the risk entirely.

Three things follow, and the third is a limitation rather than a result.

The FX leg is not intrinsically unattractive. At $\kappa = 1$ the European program buys 2.95% FX coverage. At $\kappa = 1.25$ it buys 18.31%, and residual volatility falls from 0.0916 to 0.0756, an improvement of 17.4% that no amount of extra budget can buy at $\kappa = 1$ (Section 6.1 showed the cap's shadow value exhausts at KRW 45.35bn). The 97/3 split is therefore not the statement that FX coverage is worth little. It is the statement that, when a barrel of coverage and a dollar of coverage compete for one notional envelope, the variance arithmetic prefers the barrel.

Under American premiums the constraint is the entire binding force. The two Shapley-attributed premiums sum to KRW 36.22bn at full coverage of both legs, comfortably inside the KRW 45bn authority. At $\kappa = 2$ the American program therefore buys $w_1 = w_2 = 1$ and drives σ_{res} to zero, with the budget still slack by nearly KRW 9bn. Every interior result reported for the American regime in this paper, the 0.9660/0.0340 split included, exists only because $\kappa = 1$ forbids that corner.

The consequence for how the paper's results should be read. The closed-form asymmetry (7) and the vertex geometry of Section 6.1 are correct statements about the program as specified, and the instrument-mix conclusion of Section 8, which compares two ways of buying the *same* WTI coverage, does not depend on κ at all. But the headline allocation is a joint consequence of the variance ratio and the coverage cap, and the cap carries at least as much of it as the arithmetic does. A firm whose treasury does not in fact regard the two legs as sharing one envelope, in the sense of (6), should not adopt the 0.97/0.03 split; it should re-solve at its own κ. This is a sharper caveat than the correlation sensitivity of Section 6.5, because it does not require the inputs to be mis-estimated at all.

7 The Knock-Out Survival Haircut

The American structure's premium advantage (KRW 32.0bn against KRW 41.3bn for full WTI cover) is bought with a knock-out barrier: the option dies if WTI trades outside its barriers before maturity. This creates a specific tension with the stress ledger of Section 3.2, which prices the *unhedged* fraction at the stress scenario while implicitly treating the *hedged* fraction as fully protected there. For a knock-out structure that treatment is optimistic in precisely the scenario that matters: at the stress price of 113 USD/bbl the upper barrier at 120 is close, and a knocked-out call protects nothing.

The exposure is priced with one parameter. Let p_{KO} be the probability that the structure is dead under the stress scenario. The protected fraction of each American leg is then $w(1-p_{KO})$ and not w (one WTI barrier kills both legs of the joint structure), and the stress-adjusted total cost becomes

$$\widetilde{C}^{AM}(w_1, w_2; p_{KO}) \;=\; \text{premiums} + \big(1-w_1(1-p_{KO})\big)\, UL_{WTI} + \big(1-w_2(1-p_{KO})\big)\, UL_{FX}, \quad (9)$$

where $UL_{WTI} = Q_{oil}\max(0, S^{st}_{WTI} - S_{WTI})\, S^{st}_{KRW}$ and $UL_{FX} = Q_{USD}\max(0, S^{st}_{KRW} - S_{KRW})$ are the full-exposure stress losses, KRW 105.59bn and KRW 1.48bn respectively. At $p_{KO} = 0$, (9) reduces to (3).

Figure 4 traces (9) in p_{KO}, both at the adopted risk-minimizing allocation and minimized over all feasible (w_1, w_2). Two structural facts emerge.

The budget is breached at $p^*_{KO} = 10.91\%$. While $\partial\widetilde{C}^{AM}/\partial w_1 < 0$ the cheapest attainable stress-adjusted cost is $\min_{(w_1,w_2)} \widetilde{C}^{AM} = C^{AM}(1,0) + p_{KO}\, UL_{WTI}$, which crosses the KRW 45bn cap at

$$p^*_{KO} \;=\; \frac{B - C^{AM}(1,0)}{UL_{WTI}} \;=\; \frac{45{,}000{,}000{,}000 - 33{,}477{,}827{,}504}{105{,}586{,}000{,}000} \;=\; 0.1091. \quad (10)$$

Beyond a 10.9% stress knock-out probability, *no allocation whatsoever* keeps stress-adjusted cost within the budget: the term $p_{KO} \cdot UL_{WTI}$ is a loss floor that no choice of (w_1, w_2) can hedge away, because the instrument that would hedge it is the instrument that is dying.

That formula is valid only while the coefficient on w_1 in $\widetilde{C}^{AM}$ is negative, that is while

$Q_{oil} P^{Sh}_{WTI} e^{r_w T_1} < (1 - p_{KO})\, UL_{WTI}$. The coefficient changes sign at

$$p^{\dagger}_{KO} \;=\; 1 - \frac{Q_{oil}\, P^{Sh}_{WTI}\, e^{r_w T_1}}{UL_{WTI}} \;=\; 0.6969, \tag{11}$$

above which buying WTI coverage no longer pays for itself against the survival-adjusted stress loss and the unconstrained minimiser moves from $(1, 0)$ to $(0, 0)$, where the ledger reads $UL_{WTI} + UL_{FX}$ = KRW 107.06bn. Since $p^{*}_{KO} = 0.1091$ lies well below $p^{\dagger}_{KO}$, the threshold itself is unaffected. The measured stress mortality of 0.8925, however, lies *above* $p^{\dagger}_{KO}$, so the "cheapest attainable" curve plotted in Figure 4 and the figure of KRW 127.72bn quoted for it are the $(1, 0)$ branch extended past its own validity range; the correct minimum at the measured mortality is KRW 107.06bn. Nothing downstream turns on this, because the comparison that matters is between the adopted all-KO ledger (KRW 127.15bn) and the KRW 45bn authority, and both the incorrect and the corrected minimum exceed the authority by a wide margin. The correction is recorded because the figure as printed is a minimum that a feasible point undercuts.

Measuring p_{KO}: conditioning is everything. The structure's unconditional knock-out statistics already sit well above p^{*}_{KO}. Two are in circulation and they answer different questions: 45.1% is the raw barrier-touch frequency under exact first-passage monitoring, which is the correct unconditional statistic for a survival ledger, while 23.09% is that figure net of early exercise, which understates barrier risk because an exercised path is tallied as exercised rather than knocked out. The companion paper Park (2026b) makes the same point in its Section 6.1. The survival ledger of this section needs the barrier-touch figure, but they are the wrong numbers for the stress ledger: they average over paths launched from the calm spot of 78.94, most of which never approach the barrier. The number the ledger needs is conditional on the stress state itself. Re-simulating the calibrated jump-diffusion of Section 5 from $S_{WTI} = 113$ (only 6.2% below the 120 barrier, with 32.4% diffusive volatility, 6.8 jumps per year and 0.833 years still to run) gives

$$p^{stress}_{KO} \;=\; 0.8925 \,\pm\, 0.0007 \qquad \text{(200,000 paths, jump-adapted exact first passage, asymmetric up/down jumps),} \tag{12}$$

2.0 times the unconditional barrier-touch rate and 8.2 times the infeasibility threshold (10).

Which conditional this is, and which one the ledger actually needs. Equation (12) launches a *fresh* contract from $S_{WTI} = 113$ with the full 0.833 years still to run and the barrier only 6.2% away, so it answers: if a desk were standing at the stress spot today and bought this structure, how likely is it to die? That is not the importer's question. The importer holds a contract struck and bought at 78.94 and wants to know whether that contract is still alive on the paths where WTI actually reaches the stress region. Measured that way on the same engine, conditioning the original contract on a terminal WTI in $[108, 118]$,

$$p^{\text{held}}_{KO} \;=\; 0.4901 \qquad \text{(200,000 paths, 7,992 in the stress band),} \tag{13}$$

against an unconditional 0.4374 on the same bank. The held-contract figure is the one this paper's ledger should use, and the remainder of this section is read against it; (12) is retained because it is the quantity a desk pricing a *new* structure at the stress spot would want, and because the two together bracket the range over which the conclusion is tested.

The conclusion is unchanged by the substitution, and that is the point of reporting both. At $p_{KO}^{\text{held}} = 0.4901$ the all-KO ledger reads KRW 86.09bn against the KRW 45bn authority, still nearly twice the cap; the mortality is 11.6 times the instrument-switch indifference point $\bar{p} = 4.24\%$ of Section 8 and 4.5 times the infeasibility threshold $p_{KO}^{*} = 10.91\%$. Every threshold in Sections 7 and 8 is crossed on either measure. What changes is the headline magnitude: the honest statement is that an all-KO book is dead on roughly half the paths that reach the stress region, not on nine-tenths of them. The jump calibration underlying this simulation is itself asymmetric, fitted separately as an up-jump regime ($\lambda_{up} = 2.328$/yr, $\theta_{up} = +8.04\%$, $\delta_{up} = 1.68\%$) and a down-jump regime ($\lambda_{dn} = 4.462$/yr, $\theta_{dn} = -8.75\%$, $\delta_{dn} = 2.78\%$) from the underlying return history, and not the single symmetrized pool $(\lambda, \theta_J, \delta_J)$ that the pricing engines elsewhere in this paper consume for tractability. Re-deriving (12) under that symmetrized pool instead gives 0.8937 ± 0.0007, a tenth of a percentage point away, because the 32.4% diffusive volatility dominates barrier-touch dynamics over a sub-annual horizon and the jump distribution's directional asymmetry is a second-order contributor to this specific question. On either measure the pure-KO book's arithmetic is unambiguous (Figure 4): the adopted allocation's ledger reads KRW 86.09bn at p_{KO}^{held} and KRW 127.15bn at p_{KO}^{stress}, against a KRW 45bn authority in both cases. Under the very scenario the hedge is sized against, an all-KO book is, on roughly half the paths that get there, no hedge at all.

For the pure-KO book the gap is a reporting item; Section 8 makes it a decision. The allocation program of Section 3 deliberately keeps (3), not (9), in its budget constraint: constraining on (9) would render the single-instrument program infeasible without changing any decision margin, since the floor $p_{KO}\, UL_{WTI}$ is allocation-invariant. What *can* respond to (9) is the choice of instrument, which is the degree of freedom the next section adds.

8 The Mixed Vanilla/Knock-Out Program

The haircut analysis treats the instrument as given and finds that no coverage choice can rescue an all-KO book under stress. The natural response is to stop treating it as given. This section lets the optimizer choose the instrument along with the coverage: the WTI book is split as $w_1 = w_1^V + w_1^K$ between a vanilla Black-76 leg (which survives stress) and an independently-priced KO leg (which dies with probability p_{KO}), the FX leg is priced with the vanilla Garman—Kohlhagen premium, and, decisively, the *survival-adjusted* ledger becomes the budget constraint. The KO leg's premium P_K here is *not* the Shapley-attributed WTI share (KRW 15,093.75/bbl) used to cost the American structure everywhere else in this paper: that figure is a cooperative-game marginal-contribution split of the *jointly*-priced quanto premium, valid as a cost-reporting convention for a single indivisible structure but not the market price of a standalone, independently-tradable WTI KO call. Once the WTI book is genuinely split into two separately-tradable legs, the KO leg needs its own price: an independent single-asset LSMC American KO call, priced with no FX coupling and no cost attribution on 200,000 jump-adapted exact-first-passage paths, comes to $P_K =$ KRW 17,377.11/bbl, 15.1% above the Shapley share.

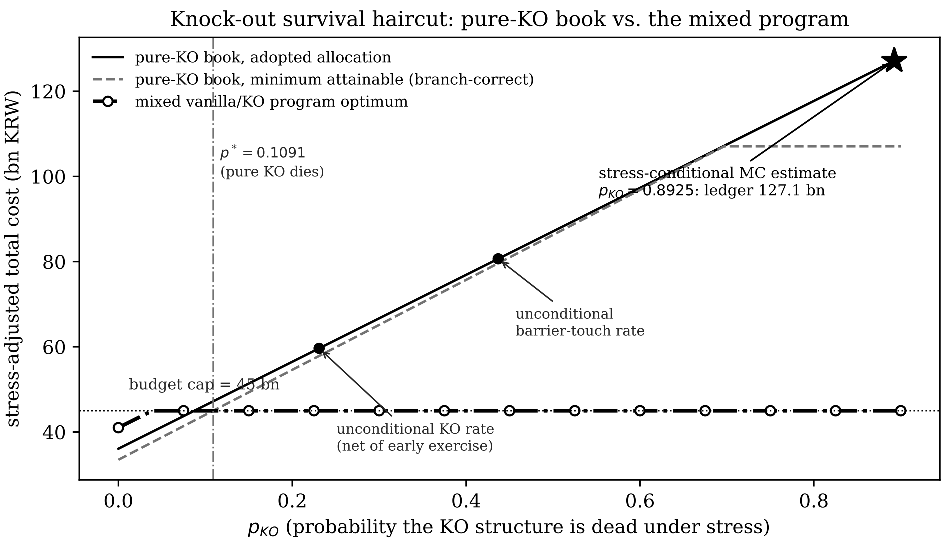

Figure 4: Stress-adjusted total cost as a function of the stress knock-out probability p_{KO}: the pure-KO book at the adopted allocation (solid) and at its cheapest attainable allocation (dashed), against the mixed vanilla/KO program of Section 8 (dash-dotted), which is ledger-feasible for every p_{KO}. The dashed curve switches branch at $p^{\dagger}_{KO} = 0.6969$ (11), above which the unconstrained minimiser moves from $(1, 0)$ to $(0, 0)$. Small markers: unconditional knock-out statistics. Star: the relaunched-contract measurement (12). The held-contract mortality (13) of 0.4901, which Section 7 argues is the figure the ledger needs, sits at KRW 86.1bn.

Both premiums in this section are cash paid at inception, so both are carried to the decision horizon on the same funding factor $(1 + r_w T_1)$ that (2) applies to the vanilla leg. Every number in this section uses that standalone price:

$$\min_{w_1^V, w_1^K, w_2} \quad \sigma_{res}(w_1^V + w_1^K,\, w_2) \quad \text{s.t.} \quad \widetilde{C}^{mix} \;\le\; B, \qquad w_1^V + w_1^K + w_2 \le 1, \quad w \ge 0, \tag{14}$$

$$\widetilde{C}^{mix} \;=\; P_V(w_1^V) + P_K(w_1^K) + P_{GK}(w_2) + \big(1 - w_1^V - w_1^K(1 - p_{KO})\big)\,UL_{WTI} + (1 - w_2)\,UL_{FX}. \tag{15}$$

Three structural facts make (14) tractable and its answer sharp.

First, the program is feasible for every p_{KO}. The all-vanilla corner $(w_1^V, w_1^K, w_2) = (1, 0, 0)$ costs KRW 42.74bn regardless of p_{KO} (vanilla protection does not die), so the infeasibility that kills the all-KO book (10) simply cannot occur. The off-ledger gap of Section 7 becomes an on-ledger constraint the optimizer can actually satisfy.

Second, the split is bang-bang, with a closed-form indifference point. At any fixed total coverage $w_1^V + w_1^K$, (15) is linear (affine) in the split, so its two marginal costs are constants, read directly off (15) by differentiating the KO- and vanilla-coverage terms separately:

$$\frac{\partial \widetilde{C}^{mix}}{\partial w_1^K} = P_K - (1 - p_{KO})\,UL_{WTI}, \qquad\qquad \frac{\partial \widetilde{C}^{mix}}{\partial w_1^V} = P_V - UL_{WTI}. \tag{16}$$

The first says: buying one more unit of KO coverage costs its premium P_K but only removes the stress loss with probability $1 - p_{KO}$, so the marginal saving is discounted by survival odds. The second says: vanilla coverage costs P_V but removes the stress loss with certainty. Since a linear program's optimal split is always a corner and not an interior mix, the two instruments are indifferent where their marginal costs cross, $\partial\widetilde{C}^{mix}/\partial w_1^K = \partial\widetilde{C}^{mix}/\partial w_1^V$:

$$P_K - (1-\bar{p})\,UL_{WTI} \;=\; P_V - UL_{WTI} \quad\Longrightarrow\quad \bar{p} \;=\; \frac{P_V - P_K}{UL_{WTI}}, \tag{17}$$

$$\bar{p} \;=\; \frac{41{,}258{,}998{,}746 - 36{,}780{,}738{,}568}{105{,}586{,}000{,}000} \;=\; 0.0424. \tag{18}$$

Below $\bar{p}$, (16)'s first expression is the smaller (more negative) of the two (KO coverage is the cheaper marginal unit, the discount outrunning its mortality), and the optimizer fills w_1^K first, so the book is all-KO; above $\bar{p}$ the ranking flips, the discount no longer pays for its own mortality, and it fills w_1^V first, so the book is all-vanilla. The optimal split is therefore an endpoint, switching at $\bar{p}$, as a linear program's corner-solution property requires; at $\bar{p}$ itself the two branch cost functions coincide at every allocation, so the switch in Figure 5 occurs at a single point. Because the standalone KO price sits well above the Shapley-attributed joint-quanto share, the switch point is low: a KO leg priced at its true independent cost is worth comparatively little mortality risk before the vanilla alternative dominates.

Third, the solution has three regimes (Figure 5). For $p_{KO} \leq 0.0390$ the all-KO branch is cheap enough on the survival-adjusted ledger that the budget goes slack and the program reaches the unconstrained variance floor $(0.9660, 0.0340)$, $\sigma_{res} = 0.0916$, the same floor the unadjusted American program of Section 6.1 attains under a KRW 45bn authority. For $0.0390 < p_{KO} < \bar{p}$ the all-KO branch is budget-pinned and its achieved risk deteriorates as mortality eats the ledger. At $\bar{p} = 0.0424$ the book switches to all-vanilla and stays there: allocation $(0.9705, 0.0295)$, survival-adjusted cost KRW 45bn, $\sigma_{res} = 0.0916$, invariant in p_{KO}. The pure-KO death threshold, recomputed with the same standalone KO price, is $p^* = 0.0638$, well above the switch: the optimizer abandons the KO leg at $\bar{p} = 0.0424$, while an all-KO book remains budget-feasible up to 0.0638. The instrument is dropped on its economics well before it becomes unaffordable, and the interval between the two thresholds is the range in which a firm could still buy the knock-out but should not.

At the measured stress mortality $p_{KO}^{stress} = 0.8925$ the verdict is not close: the mixed program selects the all-vanilla book, roughly twenty-one times $\bar{p}$ deep into the vanilla regime. Two comparisons calibrate what is gained. Against the pure-KO book at the same p_{KO}, the mixed optimum's ledger is KRW 45.00bn versus 130.67bn on the standalone-priced ledger: the entire KO-mortality exposure, KRW 86bn, priced out by the instrument switch. And against the pure-KO program's own optimum on the *unadjusted* ledger, the mixed optimum matches its residual volatility, 0.0916, while satisfying a strictly harder constraint. The instrument switch therefore costs no variance: survival-adjusted feasibility comes free. That comparison is in fact conservative, because the risk objective (1) is instrument-blind: it credits w_1 with full protection whether the WTI leg is a vanilla call or a knock-out that is dead with probability p_{KO}. Sections 7 and 8 correct for mortality on the cost ledger alone and leave the objective untouched. Carrying the same correction into the objective, by crediting the knock-out leg with effective coverage $w_1(1-p_{KO})$, raises the all-KO book's residual volatility at the measured mortality from 0.0916 to 0.372, against 0.0916 for the vanilla book. Under a survival-adjusted objective the vanilla structure is not merely equal-risk at lower survival-adjusted cost but strictly dominant on both axes; the ledger-only treatment reported here understates the case for the switch. The knock-out structure's KRW 4.48bn premium discount over the standalone vanilla leg, the only argument in its favor, is the quantity (17) says is overwhelmed once stress mortality exceeds 4.2%; the measured value is 89%, a factor of twenty-one beyond the point at which the discount stops paying for its own mortality.

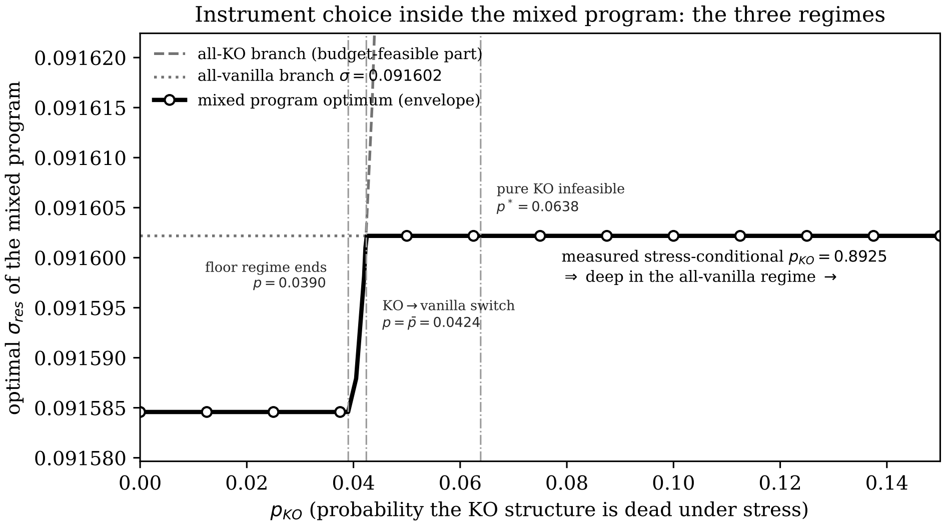

Figure 5: The mixed program's optimal residual volatility as a function of p_{KO}, with its two branches, using the independently-priced standalone KO premium. Floor regime ($p_{KO} \leq 0.0390$): the all-KO ledger leaves the budget slack and the unconstrained variance floor $\sigma_{res} = 0.091585$ is attained. Budget-pinned KO regime (0.0390—0.0424): achieved risk deteriorates as mortality consumes the ledger. All-vanilla regime ($p_{KO} \geq \bar{p} = 0.0424$): the book switches instruments and the optimum is invariant at $\sigma_{res} = 0.091602$. The measured stress-conditional $p_{KO} = 0.8925$ lies far inside the vanilla regime. The vertical axis is scaled to the 1.8×10^{-5} band the whole switch occupies, so the step at $\bar{p}$ is legible; on a conventional axis the two levels are indistinguishable, which is itself the point that the instrument switch costs essentially no variance. The all-KO branch leaves the frame immediately above $\bar{p}$.

9 Discussion

Three points deserve emphasis beyond the numerical answers.

The budget's main effect is on governance and not on risk. Section 6.1 shows the cap costs almost no residual volatility where it binds at all, yet it converts an open-ended optimization into a vertex with an auditable location: the point where the combined ledger exhausts authority and coverage exhausts the notional envelope. For a treasury committee, "we are at the corner of what you authorized" is a materially different (and more defensible) statement than "we are at an interior optimum of a model you have not seen." The shadow prices computed here give that committee the exchange rate between authority and risk: roughly half a basis point of residual volatility per additional billion KRW in the European regime, decaying to zero at KRW 45.35bn, and zero throughout in the American, where the optimal programme already fits inside the cap at KRW 36.07bn.

The asymmetry follows from the risk objective and not from the budget constraint. It is tempting to read $w_2 \approx 0.03$ as the FX hedge being "crowded out" by the budget. Equation (7) suggests a different reading: even with the budget deleted, the minimum-variance split leaves the FX leg 96.6% open. What sets the split is the variance ratio $\sigma_1^2/\sigma_2^2 \approx 18.2$ together with the low estimated correlation, both properties of the sample used for calibration and not policy choices. An importer who wants more FX coverage must argue for a different objective (for example, a tail metric on the KRW cash outflow) and not for a bigger budget.

The closed-form split inherits whatever the correlation estimate assumes. The 0.97/0.03 allocation is a consequence of (7) evaluated at a single unconditional correlation, $\rho = 0.0876$, estimated over 1,299 daily observations of a predominantly calm sample. Because the two legs are close to orthogonal at that value, the variance objective sees almost no diversification benefit from the FX leg and assigns it almost nothing. This dependence is not incidental. In stressed regimes, cross-asset correlations are widely documented to migrate sharply toward ± 1 as common liquidity and risk-appetite factors dominate idiosyncratic ones, a phenomenon usually described as correlation breakdown. Under such a migration the term $2w_1w_2\rho\sigma_1\sigma_2$ in (1) stops being negligible, the minimum-variance line rotates, and the interior split that (7) returns moves away from the corner reported here. The static single-ρ formulation is therefore best read as a calibration to normal-market co-movement; it is precisely in the joint tail, where an oil-price spike and a KRW depreciation are most likely to arrive together, that the allocation reported here has the least claim to optimality. Section 11 states the consequence.

Premium regimes should be compared on the survival-adjusted ledger, which now has a decision attached. On the naive ledger the American structure dominates: full WTI cover for KRW 32.0bn against KRW 41.3bn, an apparent saving of KRW 9.3bn. That comparison, however, prices the American leg at its Shapley-attributed share of the jointly-priced quanto structure; the independent, standalone-tradable KO instrument that the mixed program of Section 8 actually offers against a standalone vanilla leg costs KRW 36.78bn against the same KRW 41.3bn, a saving of only KRW 4.48bn. Sections 7 and 8 show that saving, whichever way it is measured, is the market price of the barrier, and that the price is currently ruinous: conditioning the knock-out probability on the stress state lifts it from the unconditional 23% to a measured 49% for the contract actually held, at which the all-KO ledger reads KRW 86.09bn against a 45bn authority. The European vanilla hedge is expensive because it survives; the American knock-out hedge is cheap because it very probably will not. The mixed program turns this from commentary into allocation: it prices the trade at a single indifference number, $\bar{p} = 4.24\%$ of stress mortality, and at roughly twenty-one times that number it walks the book to vanilla on its own.

The vanilla recommendation of Section 8 operates at a different layer than the dynamic-hedging analysis of this same structure in Park (2026b), and the two are not in tension. Such an analysis measures what happens when a desk holds the *actual* barrier contract but computes its hedge ratio from a closed-form proxy model that has no barrier term, a genuine model-instrument mismatch, since a Black-76 delta is correct for a different option (one with no barrier) and is therefore systematically wrong, in a barrier-correlated direction, on every path that approaches or crosses the knock-out boundary:

$$\Delta V_{\mathrm{KO}}(\tau_{KO}) - \delta_{B76}\,\Delta S_1 \;\neq\; 0, \tag{19}$$

the source of that analysis's order-of-magnitude hedge-cost gap and trimodal loss distribution. The instrument-mix program of Section 8 is a different decision, made earlier and at a different layer: it chooses *which contract to buy in the first place*, before any hedging begins. When the measured stress mortality selects the vanilla branch, the position that results is a genuine barrier-free WTI call, correctly hedged with the matching vanilla delta computed on that same barrier-free payoff,

$$\Delta V_{\mathrm{Vanilla}}(T) - \delta_{\mathrm{Vanilla}}\,\Delta S_1 \;\approx\; 0 \tag{20}$$

by construction, since model and instrument now agree. The dynamic-hedging warning is about pricing and hedging a knock-out contract *as if* it were vanilla; the recommendation here is about not holding the knock-out contract at all once its own survival odds are priced in. A firm that took both conclusions simultaneously would never compute a vanilla delta against a barrier position it actually holds: it would, per this paper's Section 8, have already exited the barrier position in favor of the instrument its delta model is built for.

10 Practical Implications

For the importer's hedging desk the results reduce to four actionable statements. (1) The adopted allocations are optima of a convex program, obtained from two independent solvers and checked against the dual; in the European regime a proportional 10% volatility mis-estimate leaves the allocation itself unchanged, because it preserves the variance ratio (7) reads, while a correlation shift of +0.05 relocates both regimes to (0.9770, 0.0230) with the cap slack (Section 6.5); in the American regime, pinned by the allocation line and not the cap, the same mis-estimate shifts it only from 0.9660 to 0.9770 (Section 6.5). (2) In the European regime the KRW 45bn ledger is fully committed, of which KRW 40.45bn is premium and KRW 4.55bn is the booked stress loss, and raising it to KRW 45.35bn captures the entire remaining variance benefit; in the American regime the optimal programme spends KRW 36.07bn, so the cap is not the operative constraint and further authority buys nothing. (3) The FX leg's near-zero coverage is optimal under the variance objective; if that outcome is commercially unacceptable, what must change is the objective and not the optimizer. (4) Under the measured stress knock-out probability of 49% for the contract actually held, the WTI book should be held in the vanilla instrument: the KO structure's premium discount is only worth taking if stress mortality is believed below $\bar{p} = 4.24\%$, more than an order of magnitude under the measurement, and the mixed program of Section 8 (feasible at every mortality level, budget-binding at KRW 45bn, and matching the pure-KO optimum's residual volatility) is the implementable form of that recommendation.

11 Limitations

The risk objective is a one-period standard deviation built from historical volatilities and a single correlation estimate ($\rho = 0.0876$ over 1,299 daily observations), a design choice Section 6.5 shows leaves the allocation itself invariant to proportional parameter inflation, even though the delivered risk level is not.

Before any of that, the correlation's own sampling error should be on the page, because it is larger than the paper's headline precision suggests. With $\hat{\rho} = 0.0876$ on $n = 1{,}300$ paired daily returns, $\mathrm{se}(\hat{\rho}) = (1 - \hat{\rho}^2)/\sqrt{n-1} = 0.0275$, giving a 95% interval of [0.034, 0.142]. Two consequences follow directly. The European vertex, which is the whole of question (a), requires the line solution (7) to sit inside the ledger boundary at $w_1 = 0.9705$; it crosses that boundary at $\rho = 0.1083$, which is 0.75 standard errors above the point estimate and comfortably inside the interval. And mapping the interval through (7) gives an FX coverage w_2 of [2.21%, 4.54%] against the reported 3.40%, a factor of 2.1 from end to end. The +0.05 shift examined below as a stress is 1.8 standard errors, an ordinary sampling outcome rather than a crisis scenario. Allocations quoted to four decimal places throughout this paper should be read against that interval.

That invariance result should not be read as robustness to correlation risk, which is the sharpest limitation of the exercise. Proportional inflation of both volatilities preserves the variance ratio and hence the split; a change in ρ does not. The reported allocation of 0.97/0.03 depends heavily on the low unconditional correlation estimated from a largely calm sample, and correlation is the parameter least likely to survive the stress scenario the hedge is bought against. In a crisis regime, correlations between commodity and emerging-market currency exposures can move toward +1 or −1 within days as common factors dominate; a WTI spike accompanied by a sharp KRW depreciation is the joint event a Korean importer most fears, and it is the event under which $\rho \approx 0.09$ is least defensible. Under a correlation approaching +1 the two legs become close substitutes and concentrating the premium in WTI is less costly in variance terms; under a correlation approaching −1 the FX leg acquires the diversification value the calm-sample estimate denies it, and a materially larger w_2 would be preferred. Because the objective is a one-period variance conditioned on a single point estimate, it also prices no tail risk directly: it is indifferent between distributions with equal second moments and different joint tail dependence. The 0.97/0.03 allocation should therefore be understood as optimal for the normal-market covariance structure it was calibrated on, and not as a tail-robust policy. A regime-switching or stressed-correlation covariance specification, a worst-case-ρ robust program over an interval $[\underline{\rho}, \bar{\rho}]$, or a tail objective such as conditional value-at-risk on the KRW cash outflow would each be a natural extension, and each would be expected to shift weight toward the FX leg relative to the split reported here. The cost functions price the unhedged exposure at a single stress point (WTI 113, USD/KRW 1550) and not over a distribution, which keeps them affine and their unconstrained minima corners; a distributional stress ledger would smooth the corner solutions of Section 6.3, and it would not preserve the vertex: at a stress WTI of 95 rather than 113 the European ledger at the line optimum costs KRW 43.45bn, the cap goes slack and the vertex disappears, while at a stress USD/KRW of 1700 the cheapest point in the polygon costs KRW 66.4bn and no allocation is feasible in either regime. Premium constants are treated as invariant to (w_1, w_2), consistent with the scale of this program relative to standard market depth. The stress-conditional knock-out probability of Section 7 is measured by jump-adapted exact-first-passage re-simulation from the stress state itself under the genuine asymmetric jump calibration, conditioned on a single stress spot and not a distribution of stress severities. The mixed program's standalone KO price (Section 8) is an independent single-asset LSMC valuation and not a re-attribution of a jointly-priced structure (the theoretically correct object for a genuinely split book) under the assumption that the KO leg trades on its own terms once separated from the FX leg. The static problem studied here fixes ratios once; any subsequent dynamic management of the optioned position lies outside this paper's scope. Only one stress point is examined, and the results are not stable across plausible alternatives: the vertex geometry of Section 6.1 requires a stress WTI near 113 and disappears by 95, and both the indifference point $\bar{p}$ and the infeasibility threshold p^*_{KO} scale inversely with the assumed stress loss, $\bar{p}$ moving from 23.8% at a stress WTI of 85 to 2.0% at 150. The closed-form asymmetry and the direction of the instrument-mix switch follow from the convex program itself and do not depend on the stress point; their magnitudes do. The quantitative allocation, by contrast, is conditional on the covariance inputs, and the correlation input in particular should be treated as the binding uncertainty and not as a settled parameter.

12 Conclusion

A fixed-budget importer choosing static WTI and FX hedge ratios faces a problem whose solution geometry can be mapped in closed form and verified numerically, as this paper has done. Risk minimization drives coverage to the allocation envelope in both regimes, and to the budget boundary in one: $(w_1^*, w_2^*) = (0.9705, 0.0295)$ under European vanilla premiums, committing the full KRW 45bn ledger (KRW 40.45bn of it premium), and (0.9660, 0.0340) under American knock-out premiums, spending KRW 36.07bn with the cap slack, each identified as the global optimum of a convex program by two independent solvers and a dual check. The European solution is invariant, vertex and all, under a proportional 10% inflation of both volatilities, which leaves the variance ratio and hence the line solution where they were; under a +0.05 shift in the correlation the line solution crosses the budget boundary, the cap goes slack, and both regimes relocate to the interior point (0.9770, 0.0230). Where the budget binds it does so shallowly, with a shadow price of about half a basis point of residual volatility per billion KRW and a hard floor at KRW 45.35bn. The allocation asymmetry, 0.97/0.03 in both regimes, is a closed-form consequence of an 18-to-1 variance ratio and a 0.09 correlation and not a preference of the optimizer, and it inherits that correlation estimate's calm-sample provenance: under the correlation breakdown characteristic of stressed regimes the same closed form would return a different split. Cost minimization is a genuinely different program that runs to the corner $(1, 0)$ and underspends the budget by up to a quarter, converging with the risk program only when re-anchored by the risk cap, at which point the coincidence itself indicates that the risk minima already sit on the cost—risk frontier. The American structure's premium advantage carries a mortality that this paper measures and not assumes: re-simulated from the stress state itself with jump-adapted exact first-passage monitoring on 200,000 paths under the genuine asymmetric up/down jump calibration (the symmetrized-pool re-derivation lands a tenth of a percentage point away, so the simplification the pricing engines rely on is not driving the result), the knock-out probability for a contract newly struck at the stress spot is 0.8925, and for the contract actually held it is 0.4901 conditional on WTI reaching the stress region, the latter still 4.5 times the level at which an all-KO book's survival-adjusted ledger can fit any allocation inside the budget, so the cheap hedge's protection is absent on roughly half the paths that reach the scenario it is bought against. And the mixed vanilla/knock-out program shows this is a solvable allocation problem and not a footnote: priced with an independently-derived standalone KO premium and not the jointly-priced structure's Shapley attribution, feasible at every mortality level, switching instruments bang-bang at a closed-form indifference point of 4.24%, it walks the book to all-vanilla at the measured mortality and matches the pure knock-out optimum's residual volatility under a strictly harder constraint. The static allocation question that precedes every dynamic hedging choice turns out, for this importer, to have an answer that is asymmetric, constraint-pinned, and instrument-aware, and that follows directly from the geometry once it is drawn, subject to the covariance structure on which that geometry is built.

References

[1] Park, J. (2026b). "Covariance-Aware Delta Hedging of a Knock-Out Oil Option with a Currency Leg, Under Jump-Diffusion." Working paper, Pusan National University.

[2] Adler, M., and B. Dumas (1984). "Exposure to Currency Risk: Definition and Measurement." *Financial Management* 13(2), 41—50.

[3] Anderson, R. W., and J.-P. Danthine (1981). "Cross Hedging." *Journal of Political Economy* 89(6), 1182—1196.

[4] Black, F. (1976). "The Pricing of Commodity Contracts." *Journal of Financial Economics* 3(1—2), 167—179.

[5] Ederington, L. H. (1979). "The Hedging Performance of the New Futures Markets." *Journal of Finance* 34(1), 157—170.

[6] Froot, K. A., D. S. Scharfstein, and J. C. Stein (1993). "Risk Management: Coordinating Corporate Investment and Financing Policies." *Journal of Finance* 48(5), 1629—1658.

[7] Garman, M. B., and S. W. Kohlhagen (1983). "Foreign Currency Option Values." *Journal of International Money and Finance* 2(3), 231—237.

[8] Jin, Y., and P. Jorion (2006). "Firm Value and Hedging: Evidence from U.S. Oil and Gas Producers." *Journal of Finance* 61(2), 893—919.

[9] Longstaff, F. A., and E. S. Schwartz (2001). "Valuing American Options by Simulation: A Simple Least-Squares Approach." *Review of Financial Studies* 14(1), 113—147.

[10] Markowitz, H. (1952). "Portfolio Selection." *Journal of Finance* 7(1), 77—91.

[11] Merton, R. C. (1976). "Option Pricing When Underlying Stock Returns Are Discontinuous." *Journal of Financial Economics* 3(1—2), 125—144.

[12] Reiner, E. (1992). "Quanto Mechanics." *Risk* 5(3), 59—63.

[13] Shapley, L. S. (1953). "A Value for n-Person Games." In *Contributions to the Theory of Games II*, Princeton University Press, 307—317.

[14] Stulz, R. M. (1996). "Rethinking Risk Management." *Journal of Applied Corporate Finance* 9(3), 8—25.